

HARMFUL CONTENT DETECTION

A visual system-design case study

MINIMUM SUFFICIENT DESIGN

Start with a credible baseline. Add granularity or complexity only when a requirement or measured limitation justifies it.

PRIMARY SCOPE Post-level multimodal moderation	PREFERRED MODEL Early-fusion shared-bottom multi-task model
SERVICE ACTION Keep · demote · review · remove	MAIN BASELINE Shared multi-label classifier
LABEL STRATEGY Weak reports + trusted human labels	ROLLOUT Shadow deployment first

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ML System Design Handbook · Chapter 4

0 How to read this chapter

The chapter separates source inspiration, confirmed interpretation, open decisions, and starting recommendations.

Chapter design map



Figure 0.1 - The recurring order used in this case.

FROM THE ORIGINAL NOTES

Source-derived content

Source callouts preserve the concepts and options present in the original study notes.

CONFIRMED INTERPRETATION

Clarified meaning

Teal callouts record the decisions confirmed during discussion.

OPEN DESIGN DISCUSSION

Reader decision

Amber boxes preserve genuine alternatives and provide a compact place to choose and defend an option.

STARTING RECOMMENDATION

One defensible design

Green callouts show a reasonable starting point under the current assumptions, not a universal answer.

1 Scope and service decision

Define the main service before selecting a model.

From multimodal post to moderation action

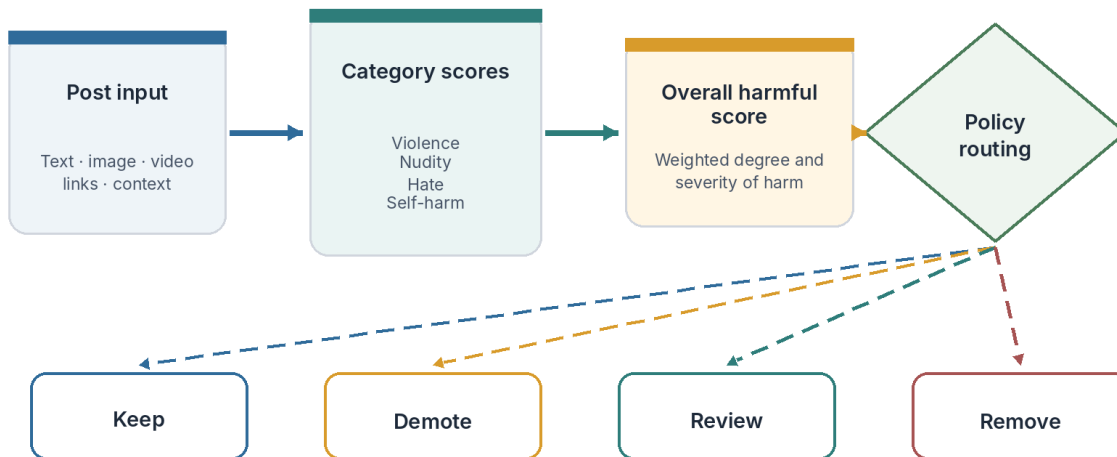


Figure 1.1 - Granular model outputs support a policy-level moderation action.

CONFIRMED INTERPRETATION

Multi-label internally, policy-driven externally

The model produces several category-level harm scores. The service aggregates them into an overall harmful score and routes the post to keep, demote, manual review, or remove.

In scope	Related but separate	Out of scope
Post-level harmful content	Bad-actor risk across behavior over time	User-facing explanation generation
Text, image, and video	Misinformation as a future case	Full actor-level architecture
Category scores and moderation action	Account-level enforcement signals	Detailed misinformation policy

DESIGN DECISION

What exactly must this service decide?

Primary unit of prediction: _____

Granular model outputs: _____

Final product action: _____

Related system boundary: _____

2 Requirements and latency

The architecture should follow the severity, modality, and operating constraints.

Requirement	Design consequence	Why it matters
Multilingual text	Use multilingual text representations	Harmful language varies across regions and languages.
Text + image + video	Use modality encoders and a fusion strategy	Meaning may depend on more than one modality.
Violence may be urgent	Maintain a real-time path	Some severe content cannot wait for a batch job.
Other categories may tolerate delay	Allow asynchronous or batch scoring	Not every category needs the most expensive path.
Policy action requires confidence	Return score and confidence	Borderline cases may need review rather than removal.

STARTING RECOMMENDATION

Split latency by severity

Use real-time inference for severe or urgent categories and asynchronous rescoring for lower-urgency categories or signals that arrive later, such as comments.

DESIGN DECISION

Which requirements actually change the architecture?

Real-time categories: _____

Batch-eligible categories: _____

Maximum online latency: _____

Required modalities: _____

3 Labels and evaluation

Use reports for scale and human review for trust.

Label strategy: scale and trust play different roles

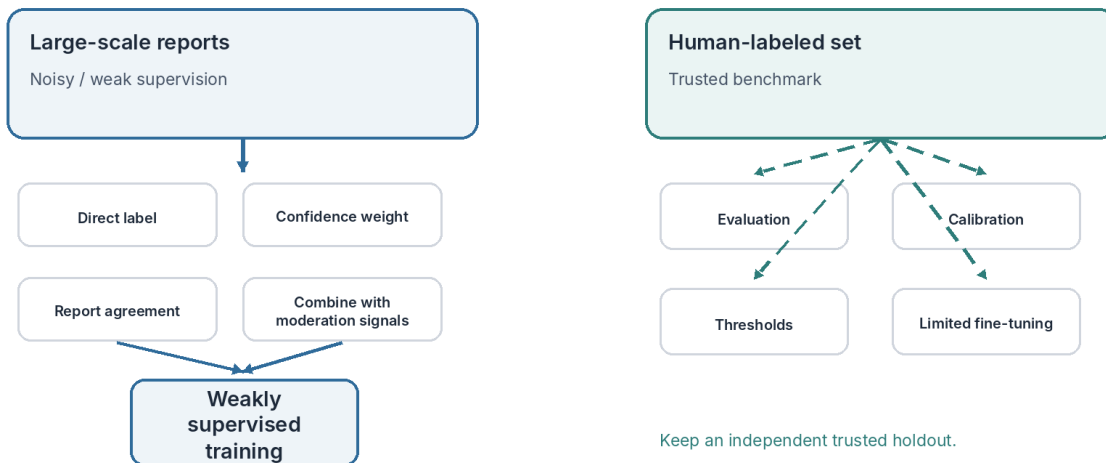


Figure 3.1 - Weak supervision and a trusted benchmark serve different purposes.

FROM THE ORIGINAL NOTES

User reports are the scalable label source

Reports reduce annotation cost, but they may be incomplete, incorrect, or malicious.

CONFIRMED INTERPRETATION

Human labels remain the trusted benchmark

Reserve carefully reviewed examples for evaluation, calibration, and threshold selection. Limited fine-tuning is possible only after keeping an independent holdout.

DESIGN DECISION

How should a user report become a training label?

Single-report treatment: _____

Multiple-report agreement: _____

Filtering rule: _____

Trusted holdout size: _____

Product and offline metrics

Metric	What it measures	Best use
Content prevalence	Harmful posts that remain visible / total posts	Volume of harmful content
Exposure prevalence	Harmful impressions / total impressions	Actual user exposure

Metric	What it measures	Best use
Valid appeal rate	Reversed decisions / appealed decisions	Moderation quality
Proactive detection	Confirmed harm found before reports / all confirmed harm	System initiative
PR-AUC	Positive-class precision-recall trade-off	Rare harmful categories
ROC-AUC	Broad score discrimination	Secondary diagnostic under imbalance

STARTING RECOMMENDATION

Choose one product metric and several guardrails

Exposure prevalence often matches user harm more directly than raw post counts. Pair it with appeal rate, proactive detection, and category-level quality.

4 Alternative 1: multimodal fusion

Compare the modular baseline with the cross-modal proposal.

Multimodal fusion: modular baseline vs cross-modal learning

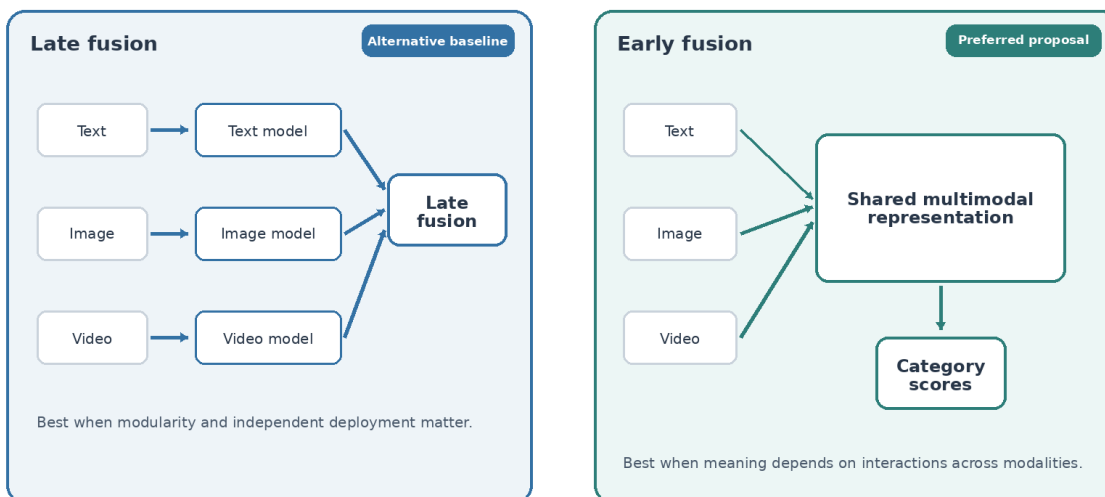


Figure 4.1 - Late fusion is operationally simple; early fusion can learn cross-modal meaning.

Option	Best when	Main benefit	Main cost
Late fusion	Modularity and independent deployment matter	Easier training, debugging, and missing-modality handling	May combine information too late
Early fusion	Meaning depends on modality interaction	Learns cross-modal relationships in one shared model	Harder training and higher compute

STARTING RECOMMENDATION**Use late fusion as the implementation baseline and early fusion as the preferred richer model**

Move to early fusion when cross-modal interactions produce measurable lift over the modular baseline.

DESIGN DECISION**What evidence justifies early fusion?**

Cross-modal failure examples: _____

Quality lift over late fusion: _____

Missing-modality behavior: _____

Latency and cost budget: _____

5 Alternative 2: prediction architecture

Increase output granularity only when the service needs it.

Classifier alternatives: from simple action to granular tasks

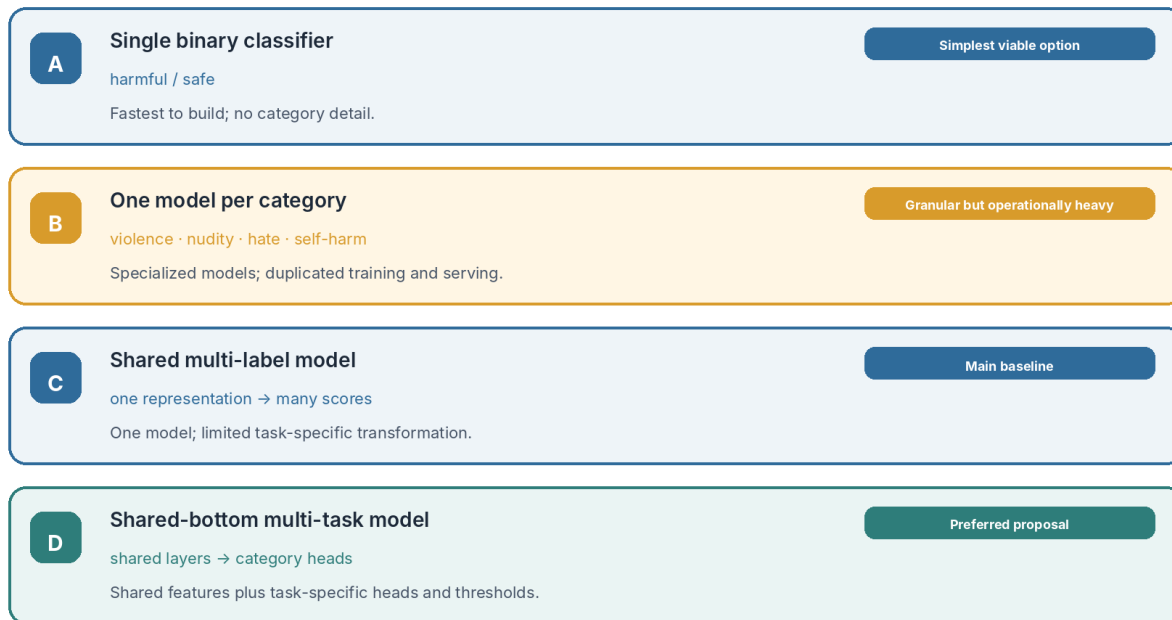


Figure 5.1 - The alternatives move from one binary action to shared, category-specific learning.

EASY TO IMPLEMENT Single binary classifier	
BEST WHEN The service only needs harmful / safe and category detail is unnecessary.	MAIN BENEFIT Fastest implementation and simplest serving.
MAIN COST No category-level analysis or severity weighting.	
GRANULAR BUT HEAVY One classifier per category	
BEST WHEN Each category needs a highly specialized model and separate operations are acceptable.	MAIN BENEFIT Strong category specialization.
MAIN COST Duplicated training, deployment, monitoring, and inference.	
MAIN BASELINE Shared multi-label classifier	
BEST WHEN One model should predict several independent categories with minimal complexity.	MAIN BENEFIT Shared representation and a single serving unit.
MAIN COST Limited task-specific transformation.	

PREFERRED PROPOSAL Shared-bottom multi-task model PRODUCTION CANDIDATE	
BEST WHEN Categories share useful features but require different transformations, losses, and thresholds.	MAIN BENEFIT Balances shared learning and category specialization.
MAIN COST More complex loss weighting and debugging.	

CONFIRMED INTERPRETATION**The service still ends in one moderation policy**

Granular heads support analysis and severity weighting. They do not require the product to expose multiple decisions to the user.

6 Representations and features

Keep model examples specific enough for implementation, but not mandatory.

Input	Illustrative encoder from the notes	Role
Text	DistilBERT, Sentence-BERT, multilingual embeddings	Language representation
Image	CLIP visual encoder or SimCLR	Visual representation
Video	VideoMoCo	Temporal visual representation
Context	Feature preparation	Posting time, device, location, selected account signals

CONFIRMED INTERPRETATION**Encoder names are examples, not requirements**

The purpose of the examples is to show the level of implementation specificity possible in an interview or internal design, while leaving room for actual benchmarking.

Missing modalities

Alternative	Benefit	Risk
Zero vector	Simplest representation	May resemble a valid low-information embedding
Special token or vector	Makes missingness explicit	Adds a learned or engineered representation
Encoder-specific handling	Fits each modality	Less uniform system behavior

Comments and author history

Feature	Simple default	Open question
Comments	Average a bounded set of comment embeddings	Use only for later rescoring, or move to attention / top-k / thread-aware aggregation
Author history	Use only information available before the current post	Include in harmful score, use for review routing, or exclude to reduce bias

DESIGN DECISION

Which features belong in the main score versus later routing?

Initial moderation features: _____

Later rescoring features: _____

Author-history role: _____

Leakage guardrail: _____

7 Overall harmful score and reference architecture

One reasonable design under the current assumptions.

One reasonable design under the current assumptions

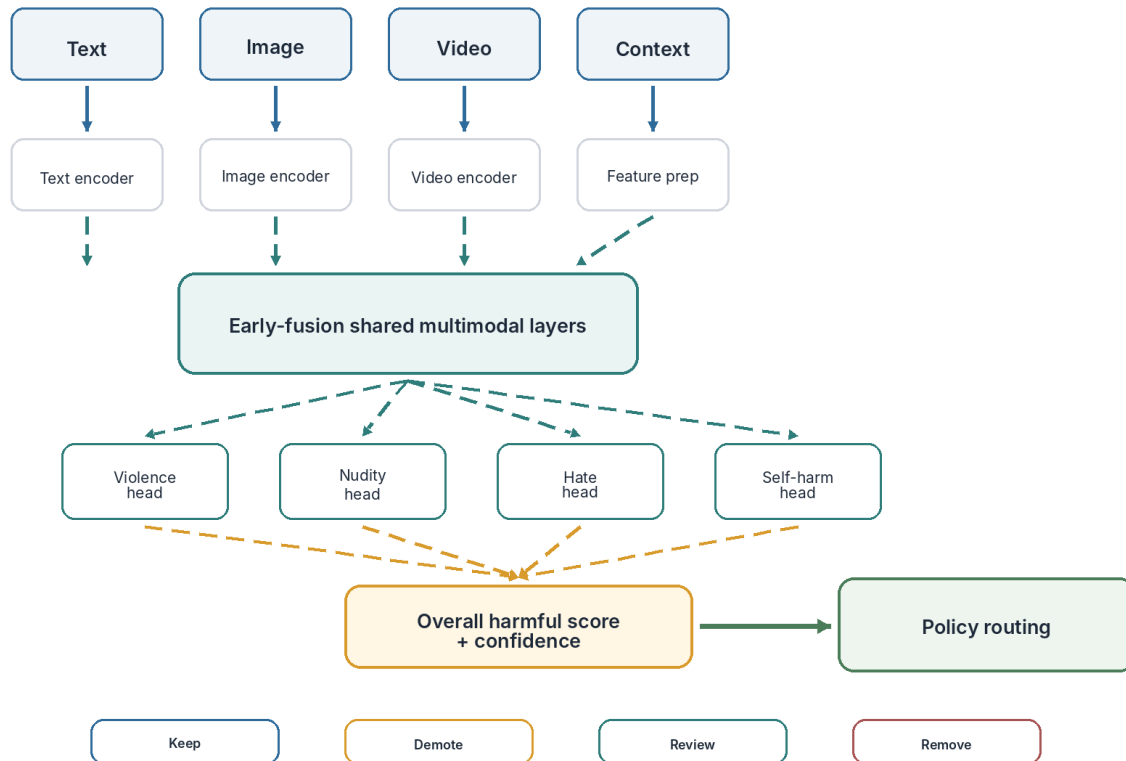


Figure 7.1 - Early fusion, category heads, score aggregation, and policy routing.

Aggregation option	Benefit	Main limitation
Weighted category combination	Transparent severity weighting	Weights require policy and validation decisions
Highest-risk category	Simple and sensitive to one severe category	Ignores combinations of moderate scores
Learned aggregation	Models interactions among categories	Less transparent and needs suitable labels
Hybrid	Combines learned score with severe-category rules	More policy and maintenance complexity

STARTING RECOMMENDATION

Use a hybrid aggregation layer

Use the learned or weighted score for general cases and explicit overrides for severe categories when policy requires them.

DESIGN DECISION

How should category scores become one harmful score?

Aggregation method: _____

Severe-category override: _____

Multiple moderate scores: _____

Calibration plan: _____

8 Training and class imbalance

Use only the methods needed by observed failure modes.

Training: balance tasks, modalities, and rare classes

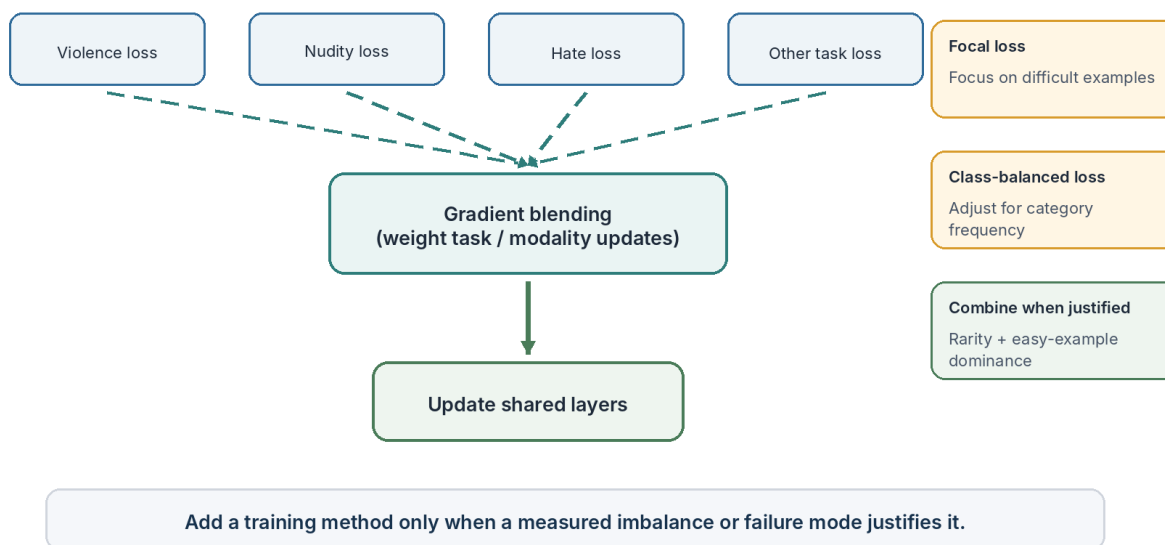


Figure 8.1 - Task losses feed a blended shared-layer update; imbalance methods solve different problems.

Method	Use when	Benefit	Risk
Gradient blending	One task or modality dominates shared updates	Balances influence on shared layers	Weights add tuning and debugging complexity
Focal loss	Easy safe examples dominate	Focuses learning on difficult examples	May emphasize noisy reports
Class-balanced loss	Category counts are highly unequal	Gives rare categories more influence	May overweight noisy rare classes
Combined loss	Both rarity and easy-example dominance are measured	Addresses two imbalance sources	More hyperparameters and interaction effects

MINIMAL-SCOPE RULE

Do not add every optimization method

Start with the simplest loss. Add gradient blending, focal loss, or class-balanced weighting only after measuring the corresponding imbalance.

DESIGN DECISION

Which training method is justified by the data?

Observed imbalance: _____

Selected method: _____

Noise concern: _____

Evidence of improvement: _____

9 Serving, latency, and deployment

Route by risk and validate silently before enforcement.

Serving and rollout: risk-sensitive paths

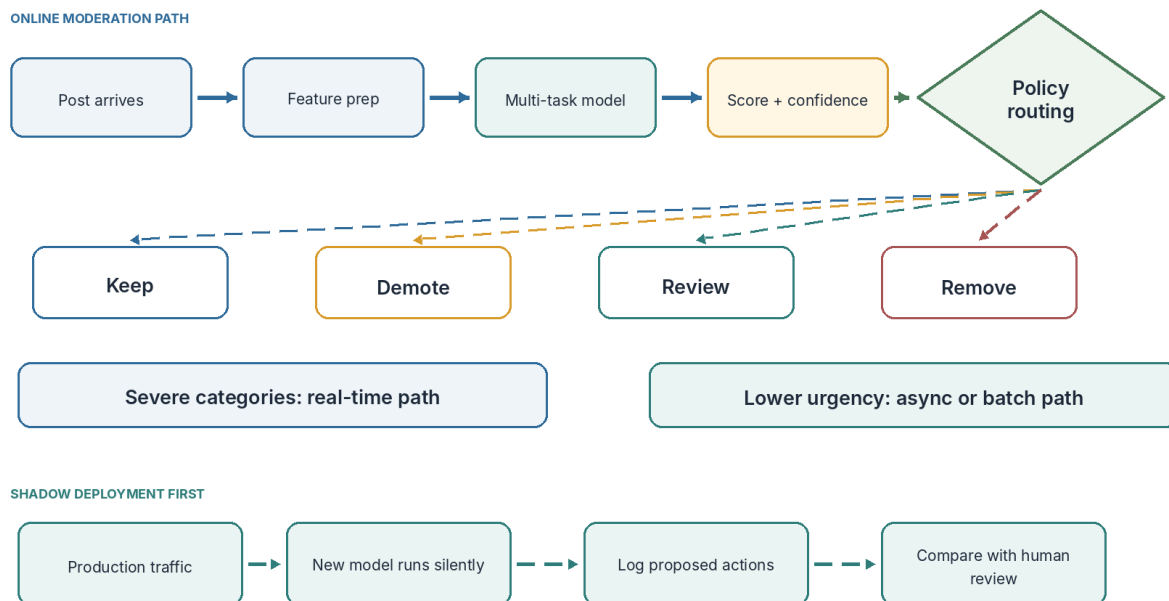


Figure 9.1 - The online moderation path and the safer first rollout step.

Score / confidence region	Action	Reason
Low risk	Keep	Avoid unnecessary intervention
Moderate risk	Demote or limit distribution	Reduce exposure without immediate removal
Borderline or uncertain	Manual review	Use human judgment where error cost is high
High risk	Remove and possibly record a violation	Act quickly on severe, confident harm

STARTING RECOMMENDATION

Shadow deployment before user-facing enforcement

Run the new model on production traffic, log its proposed actions, and compare with the current system and trusted human review.

Shadow deployment can validate	It cannot fully validate
Latency, reliability, score distribution, disagreement cases, human-review agreement	User behavior after real demotion or removal

DESIGN DECISION**What evidence permits limited enforcement?**

Shadow metrics: _____

Human-review agreement: _____

High-risk disagreement review: _____

Rollback condition: _____

10 Decision summary

A concise interview-ready reference design.

ONE REASONABLE DESIGN**Current reference architecture**

Use an early-fusion shared-bottom multi-task model, train at scale with weak report labels and evaluate on trusted human labels, aggregate category scores into a calibrated harmful score, and route content by score, confidence, category severity, and latency requirement.

Design area	Starting choice	Switch when
Fusion	Early fusion after a late-fusion baseline	Cross-modal lift is not measurable or operations dominate
Model structure	Shared-bottom multi-task	Shared multi-label performs similarly with much lower complexity
Labels	Reports for scale; human labels for trust	Report quality is too weak for useful supervision
Aggregation	Hybrid score + severe-category overrides	Policy can be represented transparently by one simpler rule
Latency	Real-time severe path; async lower urgency	All categories share the same strict latency need
Rollout	Shadow deployment first	The change is low-risk and already well validated

DESIGN DECISION**Final case decision**

Chosen design: _____

Accepted trade-off: _____

Evidence still needed: _____

Condition that changes the design: _____

Interview closing statement

INTERVIEW-READY ANSWER**Trace every component to a requirement**

I would begin with a credible shared multi-label or late-fusion baseline, measure cross-modal and category-specific failure cases, and move to the early-fusion multi-task design only when the added granularity and quality justify the training and serving complexity.